

Issues of actuarial logics of correlations in medicine as a precedent for ethics in healthcare AI

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Outline

Background

Thread 1: ML to statistics

Thread 2: Injustices of actuarial science

Thread 3: Actuarial logics in healthcare

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- I started this in 2021, after I joined Mayo Clinic, as a grounding for making an AI ethics framework, as a follow up to my 2020 preprint "A Hierarchy of Limitations in Machine Learning"
- The "framework" document supported a dense "task list" document, that proved to be too unwieldy to use. Some parts of it got out around Mayo, though
  - The "framework" document itself is also not directed enough
- I haven't worked on it since 2022; in the meantime, generative AI has taken up lots of the conversation
- Generative AI is based on an instrumental use of correlations, but we can interact with it in ways that we can't around other types of models
  - I think that generative AI will mostly recede once bills come due, but larger issues of correlations will remain. So this still worth doing, but I need help/feedback

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Maybe odd to ask this group?

- My goal is not to do original historical research (although there might be some fascinating work to do!)—I don't have the training nor inclination to attempt that—but rather use the historical through-lines to support a normative argument
- I think I have all the historical research I need to draw on for this. But it's weak in some areas (quantification in medicine; how doctors interpret probability and "risk"); have I missed something that already exists? Is there something that doesn't exist?
- *How* do I use that historical through-line support the normative argument?
- I could use (suggestions for) collaborators!

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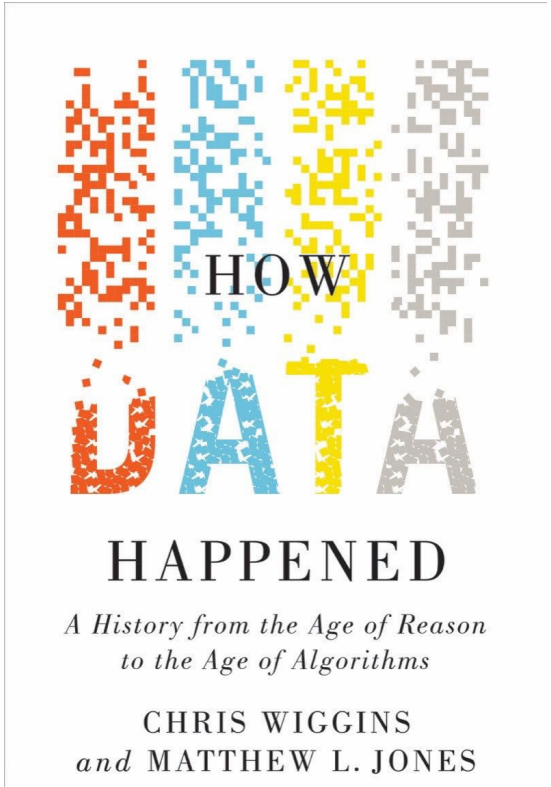
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There now is a foundational work



"Some fields, like biology, are named after the object of study; others like calculus are named after a methodology. **Artificial intelligence and machine learning, however, are named after an aspiration: the fields are defined by the goal, not the method used to get there.**"

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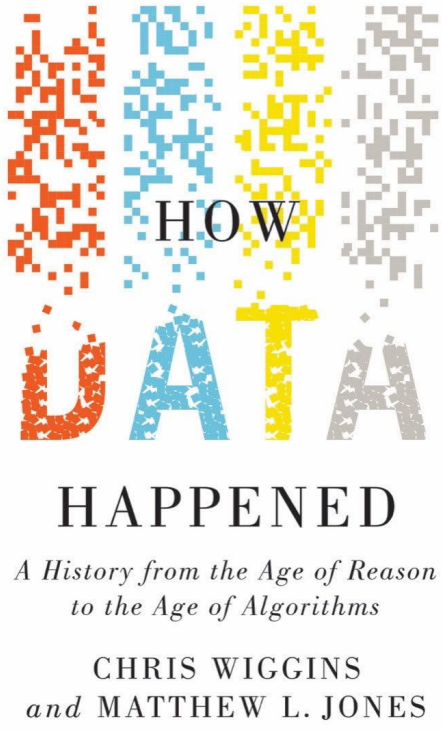
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"By the 1960s, practitioners argued, **pattern recognition succeeded in large part because it had abandoned the effort to simulate human perception**: 'Whatever successes we have had… have been the result of an effective transformation of a perception-recognition problem into a 'classification problem.' And pattern recognition researchers cared little about the symbolic side of artificial intelligence."

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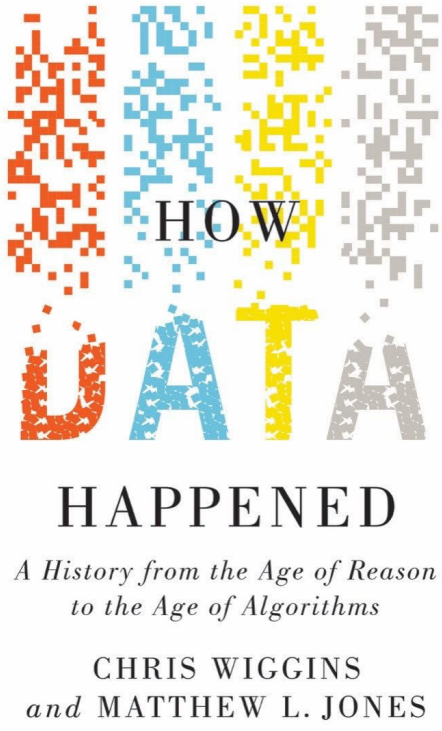
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"In a moment of profound irony, machine learning, a little-respected relative of artificial intelligence, would come in the new millennium to become the greatest success, even savior of AI, to such an extent that **after 2013 machine learning came largely to displace the far more ambitious goals of traditional AI, and the terms came to be used interchangeably**."

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"Statistical modeling: The two cultures" (Breiman, 2001)

Breiman left academic statistics for 20 years to work as a consultant, for example for the EPA (but also, unmentioned here, the military; Jones, 2018; 2023), before returning to academia. In 2001, wrote:

- "the focus in the statistical community on **data models** has:
  - "Led to irrelevant theory and questionable scientific conclusions;
  - "Kept statisticians from using more suitable **algorithmic models**;
  - "Prevented statisticians from working on exciting new problems".
- "In the past fifteen years, the growth in algorithmic modeling applications and methodology has been rapid. It has occurred **largely outside statistics in a new community—often called machine learning—that is mostly young computer scientists** (Section 7). The **advances**, particularly over the last five years, **have been startling**."

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"Statistical modeling: The two cultures" (Breiman, 2001)

- "Perhaps the damaging consequence of the insistence on data models is that **statisticians have ruled themselves out of some of the most interesting and challenging statistical problems that have arisen out of the rapidly increasing ability of computers to store and manipulate data**. These problems are increasingly present in many fields, both scientific and commercial, and solutions are being found by nonstatisticians."
- "Over the last ten years, there has been a noticeable move toward statistical work on real world problems and reaching out by statisticians toward collaborative work with other disciplines. I believe this trend will continue and, in fact, has to continue **if we are to survive as an energetic and creative field**."

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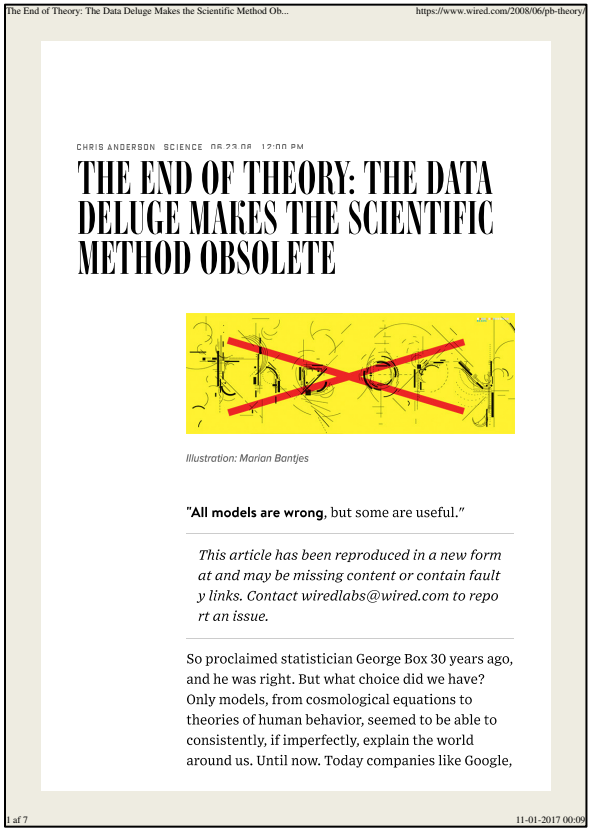
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When are correlations sufficient?



The End of Theory: The Data Deluge Makes the Scientific Method Ob... https://www.wired.com/2008/06/pb-theory/

CHRIS ANDERSON SCIENCE 06.23.08 12:00 PM

THE END OF THEORY: THE DATA DELUGE MAKES THE SCIENTIFIC METHOD OBSOLETE

Illustration: Marian Bantjes

"**All models are wrong**, but some are useful."

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So proclaimed statistician George Box 30 years ago, and he was right. But what choice did we have? Only models, from cosmological equations to theories of human behavior, seemed to be able to consistently, if imperfectly, explain the world around us. Until now. Today companies like Google,

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- In 2008, in "End of Theory", Wired's Chris Anderson argued that when data are big enough, causality doesn't matter, and correlations alone are sufficient (and we don't need "models").
  - He was trivially wrong (see Harford, 2014; Meng, 2018: and anyway, correlation is a model)
- Breiman's paper provided a legitimate version of this argument much earlier: sometimes, for some use cases, correlations alone are sufficient (and this has nothing to do with the size of the data). Then, we can (and should?) use "algorithmic" models

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Problems with existing definitions

"Artificial intelligence is that activity devoted to making machines intelligent, and intelligence is that quality that enables an entity to function appropriately and with foresight in its environment." (Nils J. Nilsson, quoted in Stanford AI100, 2016)

- Definition based on *aspiration*, not substance

"Artificial Intelligence (AI) is a science and a set of computational technologies that are inspired by—but typically operate quite differently from—the ways people use their nervous systems and bodies to sense, learn, reason, and take action." (Stanford AI100, 2016)

- Defines by what it *appears* to do, not *how* it does so

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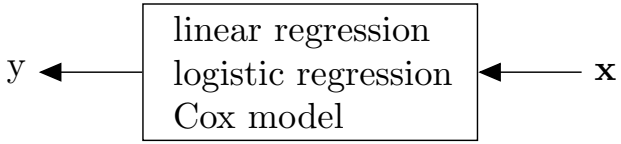
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Breiman's diagram provides a practical definition of AI/ML

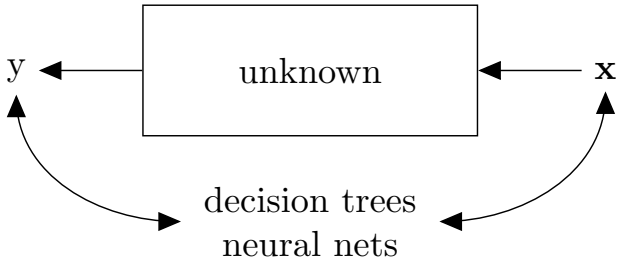
Imagined world



Statistical approach



Machine learning approach



- In practice, all the horror stories, all the success stories, are exclusively due to *statistical machine learning*
- This is an instrumental use of correlations for the purpose of "prediction" (perhaps better called "outcome reasoning": Baiocchi & Radu, 2021), that Leo Breiman theorized in 2001 with this diagram (see also, Wiggins & Jones, 2023)

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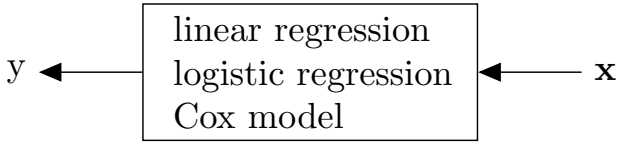
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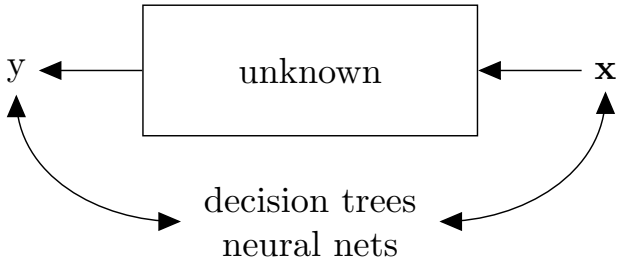
Imagined world



Statistical approach



Machine learning approach



- We can define—or at least identify—the target of concern: *the instrumental use of correlations*
- It would be better to just say "statistical machine learning", or something else (the "learning" is an example of a "wishful mnemonic"; McDermott, 1976), but we seem to be stuck with the rhetoric of "AI" (and "algorithms", even though it's all models that are the issue)

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Key question

- *When are correlations alone enough?*
- Attempts to answer after Breiman (2001) via Kleinberg et al., 2014; Mullainathan & Spiess, 2017; but those don't include an ethical lens (see Wang, Kapoor et al., 2023)
- Some interesting discussions in a 2021 special issue of *Observational Studies* for the 20th anniversary of Breiman's article
- Liu, Raji et al. (2026) might finally have a good formal framework as of this year; I am still working through it

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Thread 2: Historical injustices of correlations in actuarial science

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Historical episode of correlations being rejected

- In 1884, Massachusetts state senator Julius C. Chappelle (c. 1852–1904) introduced a bill to ban discrimination in life insurance rates on the basis of race (Bouk, 2015)
- Opponents argued that *the empirical fact of* shorter life spans for Black people, and not "discrimination on the ground of color", was what motivated insurance rates.
- It was simply "a matter of business", and it was "inexpedient to legislate" against racial discrimination. Doing so "would probably cut off insurance entirely from the colored race."



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Historical episode of correlations being rejected

- The bill passed, under two arguments:
  - Prior life spans were based on enslavement, and did not consider future possibilities under freedom
  - Setting rates based on prior life spans did not allow for the *potential* for equality
- The John Hancock company halted its discriminatory pricing strategy… but also forbade agents from soliciting from Black people and refused commissions for agents who did (effectively creating out of spite rather than necessity the reality that they had threatened, of cutting off life insurance access)



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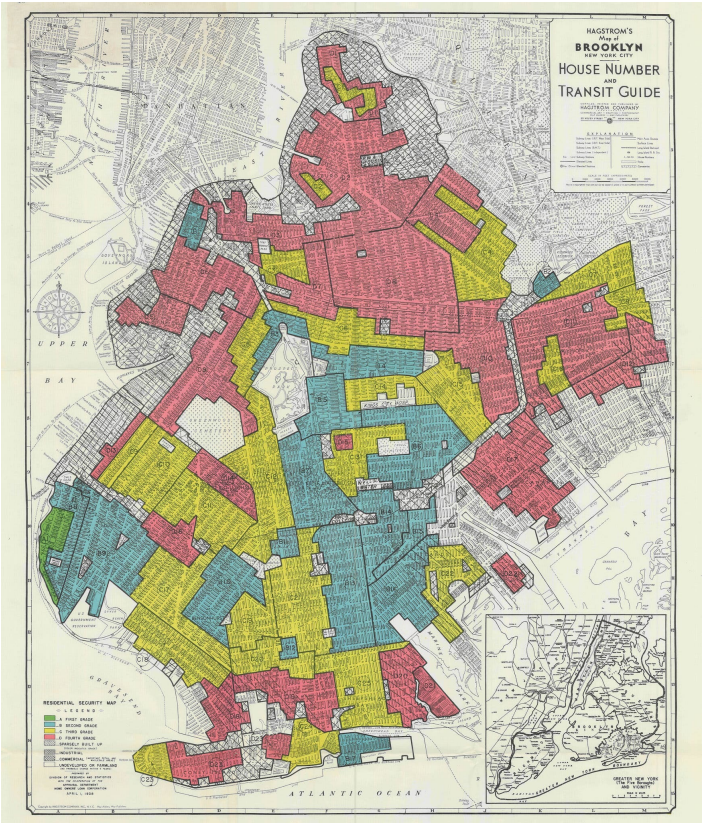
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Correlations used to justify racism



- "Redlining" originally referred to *insurance companies* putting "red lines" on maps for areas "too risky" to insure, in the 1940s and 1950s (after insurance deregulation) (Horan, 2021)
- (*Later,* the term came to be applied to federal programs like the Federal Housing Authority, which relied on color-coded maps. Constant wealth redistribution away from Black people.)
- Businesses couldn't get money to grow, which contributed to "urban blight", and that plus police violence led to uprisings in the mid-1960s

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Civil rights activist response, and industry response



When you invest a billion dollars to help the cities, you learn some things.

Like hope.

Troubled minds. Back some 18 months ago, there were a lot of troubled minds all over this country. Including many in our business. The life insurance business. And what troubled everyone was the cities. There was poverty and frustration and decay and much ugliness all the way around.

In that atmosphere, when there was precious little hope anyplace, a lot of companies from our business got together to do something about it. To give it a try. You can't ride out the ups and downs of business without a certain inherent strain of optimism.

The life insurance business decided to invest a billion dollars worth of loan funds in the city core areas. Money that would create more jobs. More housing. Hopefully, more hope. And we made a public pledge of this investment.

You may say, this was just business as usual.

Because historically, life insurance companies invest in housing and in enterprise that makes jobs.

But this was different.

This was a new and special case of investment.

It went to an area—the inner cities—where capital was not readily available on reasonable terms, because of risk and location. Our business felt this special commitment was essential.

After all, our business is totally bound up with the health and safety of people. And people live in the cities. You could say people are the cities.

If those cities crumble, people are going to crumble, and business—ours, yours, anyones—is apt to crumble right along with them.

In a businesslike way, our business was investing in its own future.

Due to the nature of the problem, the life insurance companies would need the closest cooperation of government and responsible leaders of the community. And they're getting it. With the result that the billion is now almost completely committed.

What we learned, was people.

By our very involvement in the core areas, we of the life insurance business found ourselves getting a lot closer to where people live.

We found that despite all the talk of backlash and Blacklash, there's an even greater drive to work together. A drive to create, not to hate.

We confirmed a deep feeling. That the problem of the cities needs people—people in government, business, and labor, working together—to help solve it. And we discovered we weren't trying alone; other businesses were making special efforts.

The life insurance companies are re-learning a basic truth. Let everyone do what he does best. We ourselves know investment in housing and enterprise. Local planners, developers, and agencies know their communities and know their needs.

Our business has learned that its hope was justified.

Sure, minds are still troubled today. The situation won't "just go away." But...

The life insurance business has so much hope that it's taking another step.

A second billion.

A second billion devoted to the same aims.

While knowing that it doesn't nearly fill the whole need, the life insurance business regards this, like the first billion, as an investment in its own future.

Isn't it your future, too?

If you would like some suggestions on what you can do, write for the booklet "The cities...your challenge, too."

Institute of Life Insurance
On behalf of the life insurance companies in America
277 Park Avenue
New York, N.Y. 10017

A social services center grows in Bedford-Stuyvesant, being built with mortgage money provided by the life insurance companies. One of hundreds of projects in 222 cities.

- Twenty of the largest civil rights organizations boycotted Metropolitan Life Insurance in 1966. Some called for the end of private insurance, advocating that it be nationalized, to share the risk across society rather than concentrating prosperity in white areas and risk in Black areas (Horan, 2021)
  - Future Justice Thurgood Marshal had his own experience of being denied car insurance, ostensibly because of "traffic congestion" in Harlem (and not race)
- The Urban Investment Program was an industry effort to forestall this, with insurance companies committing to invest $1 billion in urban communities in 1967 with **lots of publicity**

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- But companies worried that other (white) policyholders would object, and so insisted that this would not be a "giveaway" but at "market rates." Lack of multiple types of insurance (housing, fire) made those areas "riskier," and market rates high
- Vague standards for what would "stabilize" communities or provide services to residents let companies **invest the money in more profitable, white-owned factories and warehouses; white-owned and white-employing shopping centers at the edges of town; and white-owned medical facilities which employed but not served "residents of the urban core"**
- Another $1b commitment in 1968, but with fewer participants, and quietly discontinued in 1972

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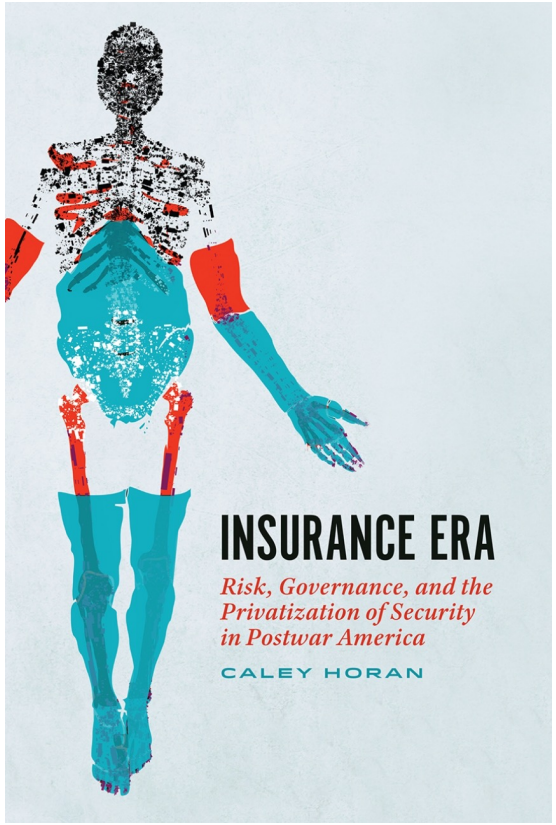
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The "unisex insurance debate"



- Sex was considered a risk factor for health insurance, leading to women being charged more (although less for life insurance). Unlike race, which was indirect via historic patterns of segregation and ownership, sex was used overtly and openly
- 1960s and 1970s feminist campaigners sought to combat this discrimination
- Insurance companies ultimately triumphed, defeating proposed legislation in the 1980s through inventing the idea of "actuarial fairness", which defended practices in "objective, statistical terms, and bolstered their case with market-based logics and an ideological claim to fairness" (Horan, 2021)
- Later on, this led to health and life insurance starting in the late 80s to charge more to women who were the victims of domestic violence

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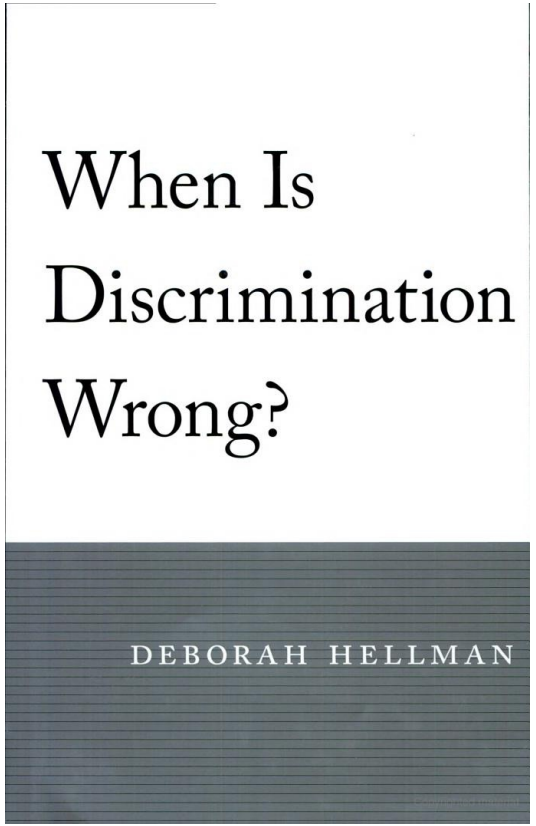
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Consequences of sex-differentiated rates

- Charging higher rates for health and life insurance for women who were the subject of intimate partner violence was "rational" for the insurance industry, as "**women who are being beaten by their husbands or boyfriends are more likely than the average insurance purchasers to be hurt or even killed.**"
- Insurance did not distinguish between survivors who had left and those who didn't, because leaving is the time of greatest danger



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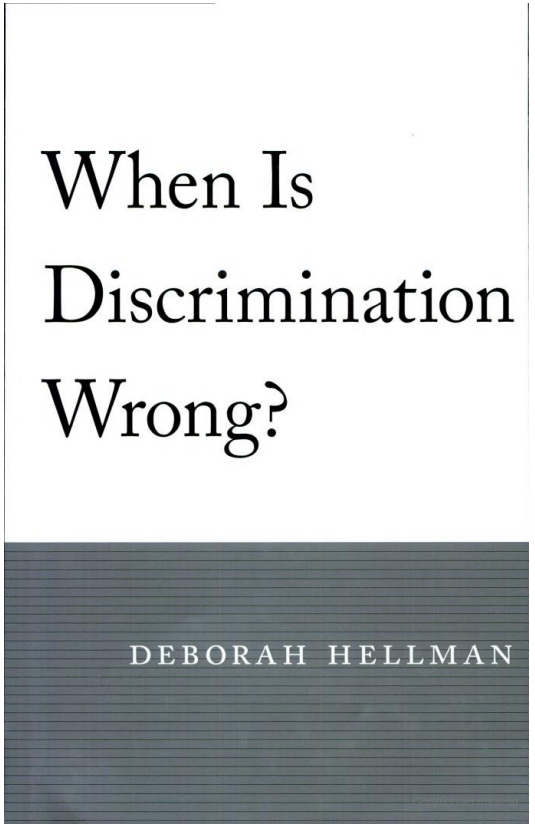
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Consequences of sex-differentiated rates

- "In my view, which I hope is shared, **this practice is wrong**. Insurers ought not to use a person's status as a victim of domestic abuse as a criterion in determining whether to insure her or at what rate. This is so notwithstanding the fact that the generalization underlying the discrimination is accurate… [which] suggest[s] that **accuracy is not causally related to the permissibility of classifications**."
- "**Rather, if anything, it highlights the need to not pile on by refusing to sell insurance to victims of injustice and cruelty**." (Hellman, 2008; see also Hellman, 2021)



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The problem with "actuarial fairness"

- "Actuarial fairness" was a concept invented in the 1970s in response to these activist campaigns
- This is: if it correlates, it is "fair" to use. This moved the debate into a technical realm, a move which helped secure victory (Horan, 2021) with exceptions for explicitly using race or gender
  - This has direct parallels to "algorithmic fairness" (covered later, Ochigame, 2019)
- People find the *outcomes* of actuarial fairness deeply unfair (Kiviat, 2021), with many moral arguments against actuarial fairness (Simon, 1988; Landes, 2015; Heras et al., 2019; Frezal & Barry, 2020)
- The actuarial science industry is always trying to get more things allowed in calculating "risk", because that helps them avoid risky individuals and thereby keep as much money as possible
  - The CEO of a company using unconventional data for lending decisions: "all data is credit data" (Kiviat, 2019)
- Key input into the impacts of classification (e.g., Fourcade & Healy, 2010)

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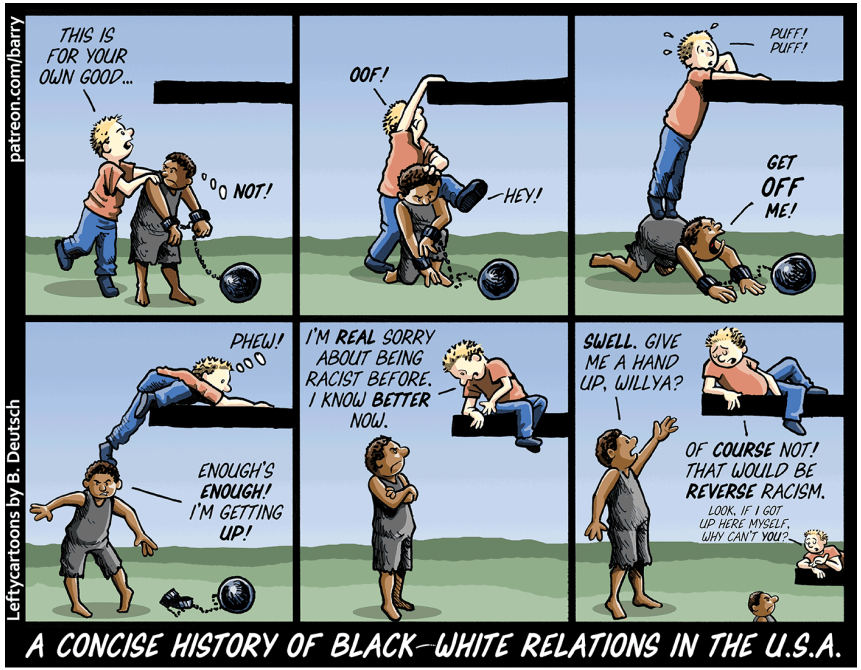
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Anticlassification vs antisubordination

- Anticlassification: "prohibits practices that classify people on the basis of a forbidden category" (including for the purpose of challenging group subordination). This is what employment law has changed into (Arehart, 2012)
- Antisubordination: "guarantees of equal citizenship cannot be realized under conditions of pervasive social stratification and argue that law should reform institutions and practices that enforce the secondary social status of historically oppressed groups." (Balkin & Siegel, 2004)



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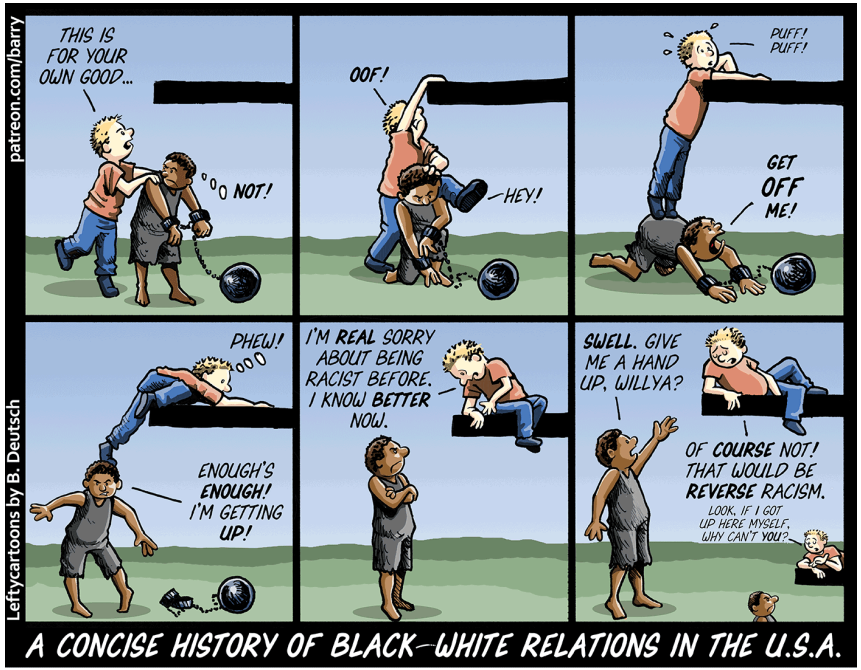
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Anticlassification vs antisubordination

- Anticlassification: "The way to stop discrimination on the basis of race is to stop discriminating on the basis of race" (Roberts' opinion in 2007 *Parents Involved* SCOUTS decision). Critiqued as acontextual and ahistorical (Turner, 2015)
- Antisubordination would change this to, "the way to fix inequalities in the category of race is to fix inequalities in the category of race" (Hu, 2019)



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What still exists today



- ZIP Code is a legal basis for determining car insurance rates!
- Fergus (2013) calls this a "ghetto tax", because of how the history of segregation and redlining means that Black and brown folks have been forced into certain areas, areas that are subject to disinvestment, environmental racism, and a lack of infrastructure.
- *Even if* living in these places leads to more damage to cars regardless of individual driving, *is this something that individuals should be held responsible for?* Insurance companies say yes, and so far the government has agreed
- Recent (Huang & Hooker, 2026) econometrics estimate: minority ZIP codes in Illinois pay $34–$158 more per year

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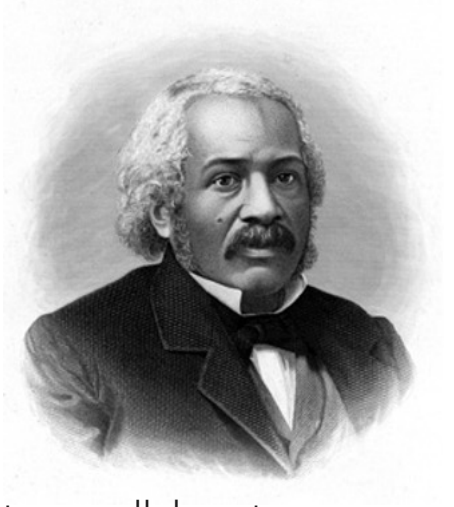
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Points of quantification in medicine



Recent AmStatNews post (Reynolds, 2026):

- "1840: James McCune Smith was the first African American to obtain a medical degree and was a founding member of the New York Statistical Society. In 1840, he published a series of articles in a leading medical journal demonstrating that silver nitrate treatments for sexually transmitted disease in women were ineffective and harmful. Later, he used statistical arguments to debunk bogus and pseudoscientific medical practices such as phenology and the "lay puffery" of homeopathy.
- "1860: At the fourth International Statistical Congress in London, long-term collaborators Florence Nightingale and William Farr presented a groundbreaking proposal for standardizing hospital statistics, establishing a foundational framework for modern hospital quality improvement and patient safety and identifying corrective actions for reducing harm to patients and healthcare providers."
- 1861: Handwashing. "Using the 'numerical method,' advocated by Pierre-Charles Alexandre Louis, Ignaz Semmelweis published a quantitative comparison of mortality rates between two maternity wards at Vienna General Hospital."

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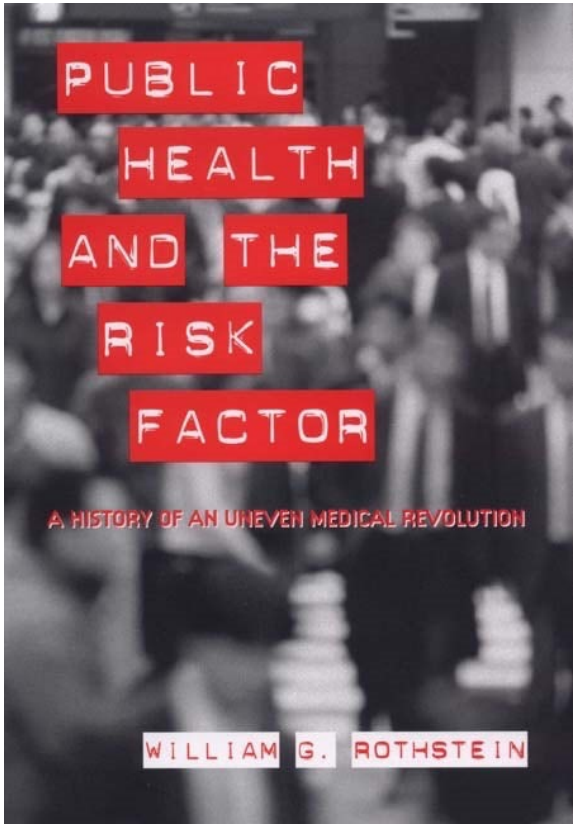
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Correlations first came into medicine in force via insurance



"The life insurance industry readily accepted 'build' because it was **interested in predictors of mortality, not causes, and especially predictors that were inexpensive, accurate, consistent, and simple to measure**. By contrast, **most medical scientists insisted that predictors be incorporated into causal models and tested experimentally with laboratory animals, as was being done with bacterial pathogens**… [In 1890] Edward Trudeau (1848–1915), one of the first Americans to investigate the tubercle bacillus… diagnosed tuberculosis in a young man who insisted that he was in perfect health because he had just been approved for policies by medical examiners from two leading life insurance companies."

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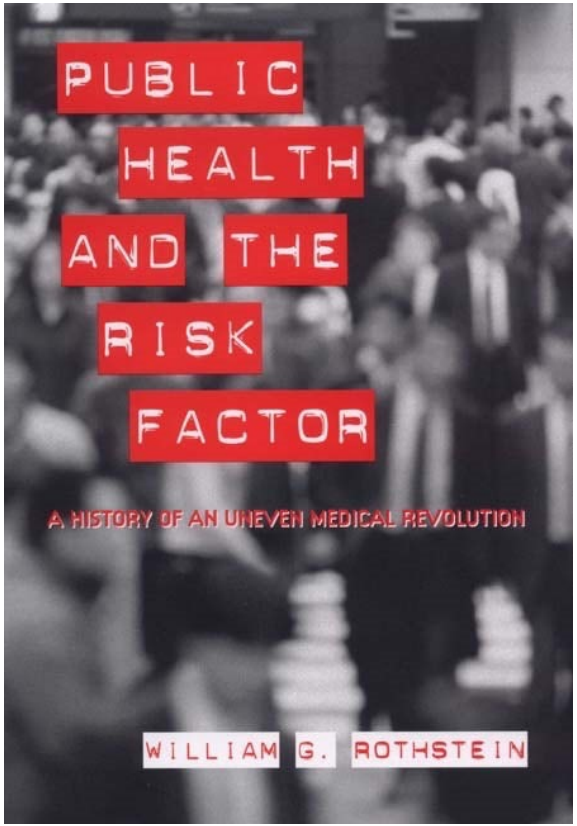
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Correlations first came into medicine in force via insurance



"As life insurance companies discovered more risk factors and accumulated statistics on each of them, the role of the medical examiners changed significantly. Companies increasingly required examiners simply to send their findings, together with urine specimens that required laboratory analyses, to the home office. In 1915, the president of the Association of Life Insurance Medical Directors of America noted with disapproval in his presidential address… **Formerly the physician evaluated the applicant; now the physician provided the data and the company made the decisions based on statistical expertise that the physician did not possess**."

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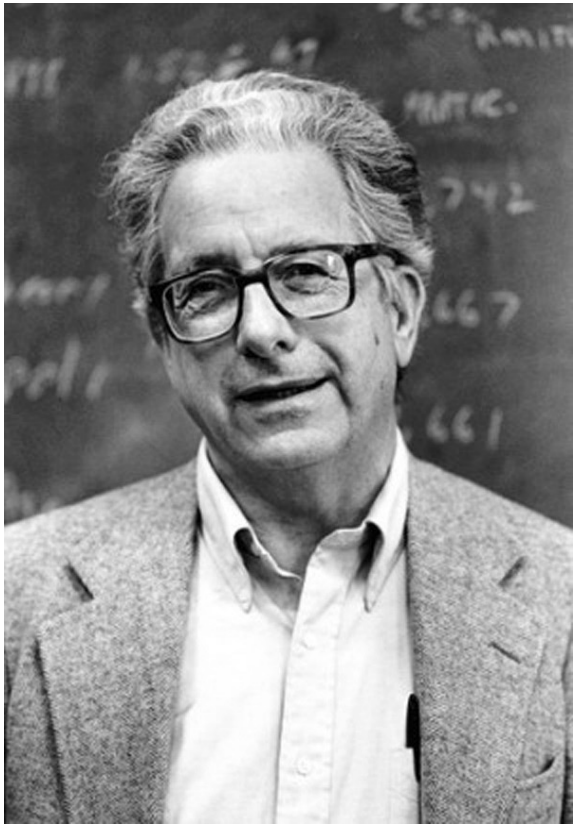
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Mid-20th century: Paul Meier (1924-2011)

- "In those early days [late 1940s], Meier said, 'I was looked at with amazement by my medical colleagues,' when he brought up the idea of randomization for assessing new medical treatments, he recalled. The physicians would say "'Randomize? We know that this treatment is better than that one,"' he explained. 'People who knew and respected me were astounded that I should want to randomize their patients.'" (Betts, 2012)
- "The Society for Clinical Trials called the [1954] polio vaccine trial 'the project that put randomized trials on the map in this country' in part because of the key role Meier played by publishing a critical article in *Science* in 1957." (Ibid.)



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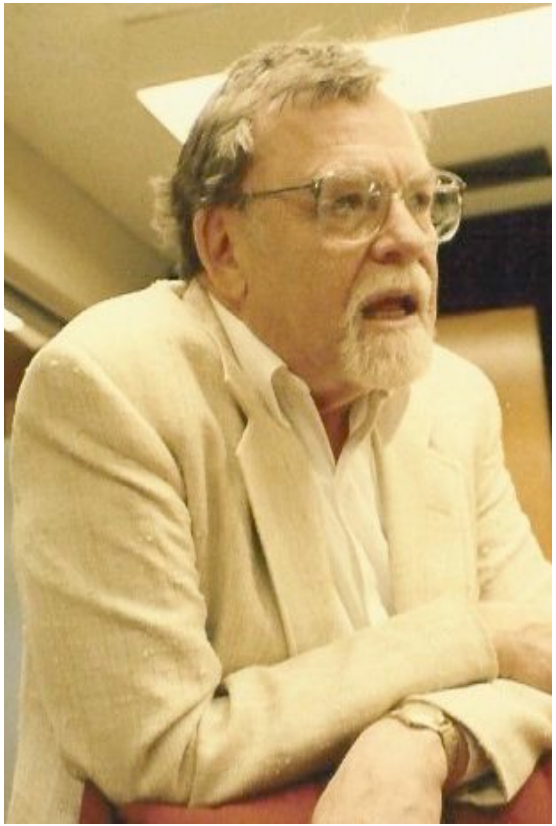
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Mid-20th century: Paul Mehl (1920-2003)



- "The mechanical combining of information for classification purposes, and the resultant probability figure which is an empirically determined relative frequency, are the characteristics that define the actuarial or statistical type of prediction. Alternatively, we may proceed on what seems, at least, to be a very different path. On the basis of interview impressions, other data from the history and possibility of psychometric information of the same type as in the first type of prediction, we formulate, as in psychiatric staff conference, some psychological hypotheses regarding the structure and dynamics of this particular individual…. This type of procedure has been loosely called the clinical or case study method of prediction." (1954)
- Argued for actuarial over clinical prediction in psychology; I think this has a big impact?

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Other 20th century developments

- Framingham Heart Study led to the modern term/idea of risk factors (Rothstein, 2003)
  - Rothstein argues it still sits uneasily within medicine, which I think applies to AI/ML as well
  - There is a logical gap between probability statements and what can be applied to any given individual (Dawid, 2012; Stark & Freedman, 2003; Faigman et al., 2014). Doctors don't try to make decisions that will "on average" lead to best outcomes; they should try for the best outcome for an individual patient
- Evidence-based medicine from the 80s onwards; increasing documentation and reporting demands made on medical professionals; that, plus more patients, led to less interaction time
- Critiques of neoliberal artificial scarcity and "lifeboat ethics" (Koch, 2012), and of impersonal, industrialized models of care (Montori, 2020)

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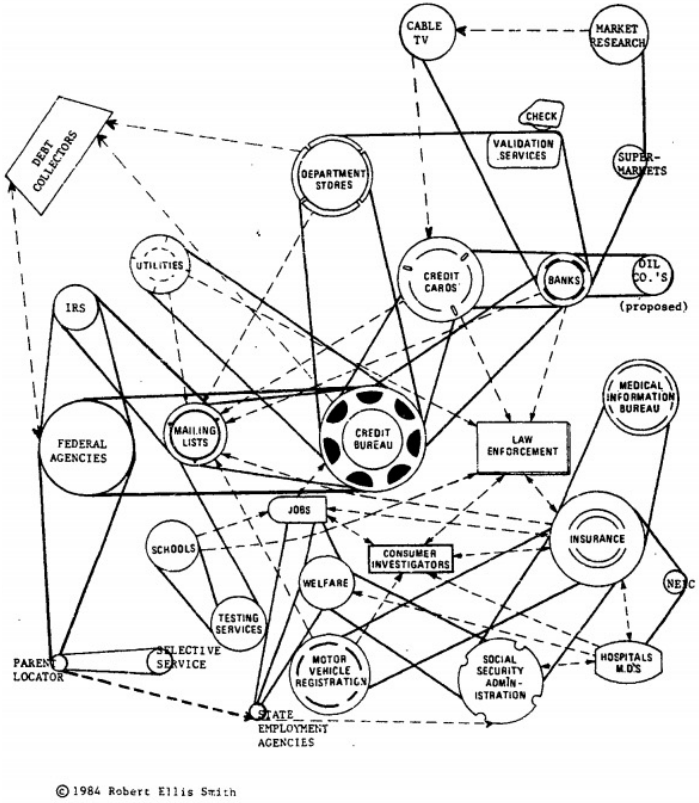
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Thread 1 + 2: Insurance/Injustice and AI

- This entire work was inspired by Rodrigo Ochigame's "The Long History of Algorithmic Fairness" (2020)
- Done while they were a PhD student in MIT's HASTS program and embedded in Joi Ito's lab (see also Rodrigo's expose in *The Intercept* from that experience), they presented to a seminar with Caley Horan, who said, "we need to talk!"
- The "algorithmic fairness" debates of today directly resemble the "actuarial fairness" debates that is the topic of Horan's 2019 dissertation (and 2021 book), in ways that hadn't been acknowledged!
- Since then, some of work from Deborah Hellman and others have touched on the legal theory/philosophical overlap, but don't necessarily have the angle of social movements and voices of those who are impacted (conversely, the work that has those doesn't connect to this specific history)



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More than a resemblance; a convergence

Bruce Hoadley (2001), respondent to Breiman's paper:

- "[My] last seven years have been with the research group at Fair, Isaac. This company provides all kinds of decision support solutions to several industries, and is very well known for credit scoring. Credit scoring is a great example of the problem discussed by Professor Breiman."
  - Hoadley describes the independent (and less sophisticated) invention within credit scoring of several techniques from machine learning, including decision trees, later eventually adopting machine learning techniques directly
- "These early Fair, Isaac engineers were ahead of their time and charter members of the algorithmic modeling culture."
  - Breiman's rejoinder: "They sound like people I would enjoy working with."

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Antisubordination in machine learning?

- Some papers making the connections (Ho & Xiang, 2020; Simons et al., 2021; Keswani & Celis, 2022)
- Practically, this may be difficult to implement; antisubordination requires resource redistribution, and how much we ought to redistribute, to whom, is a moral question that cannot be answered from existing patterns
- We can try to quantitatively model historical counterfactuals, but this is hard if not impossible to do meaningfully do quantitatively because we never have data from a counterfactual state of the world without oppression.

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Wang, Kapoor, et al., "Against Predictive Optimization" (2023)

- (I was a reviewer for an early version of this paper)
- Critiques seven types of mismatch between models based on optimization, and decision-making:
  - "Good predictions may not lead to good decisions (intervention vs. prediction)
  - "It's hard to measure what we truly care about (target-construct mismatch)
  - …
  - "Providing adequate contestability undercuts putative efficiency benefits (Lack of contestability)
  - "Predictive optimization doesn't account for strategic behavior (Goodhart's Law)"

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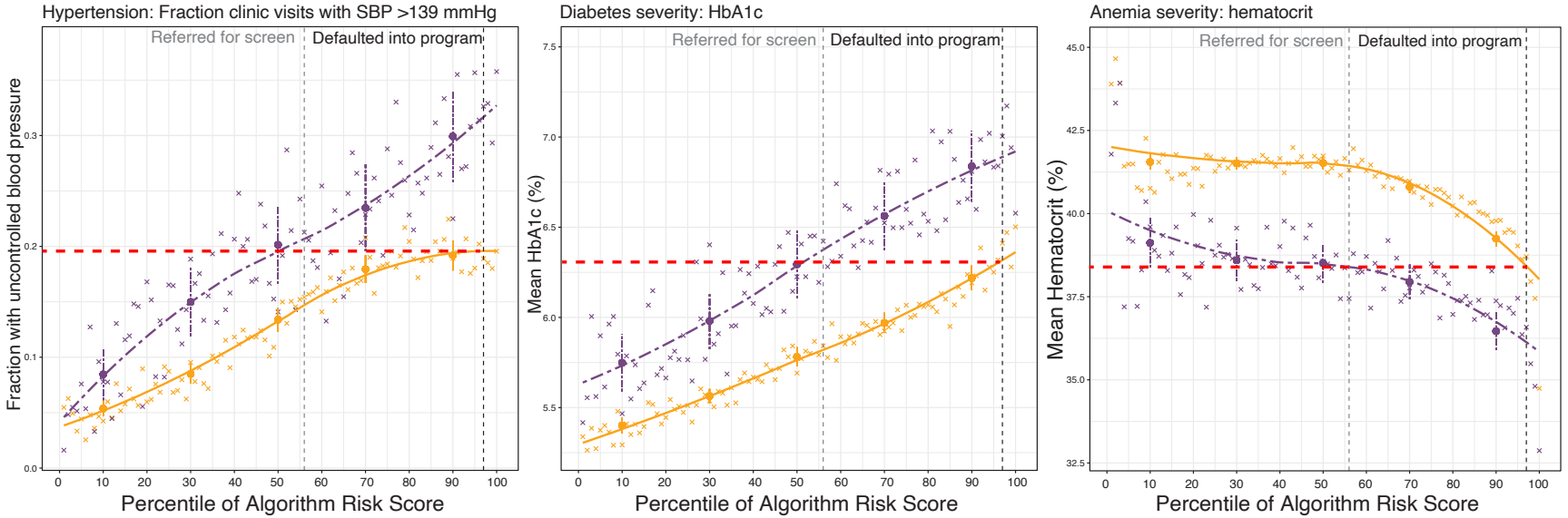
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Thread 2 + 3: Healthcare and injustice



In an ML system implemented in a major healthcare system (Optum), Black patients not even referred for screening at the point (red dashed line, added) where equally sick white patients are defaulted into programs! (Obermeyer et al., 2019)

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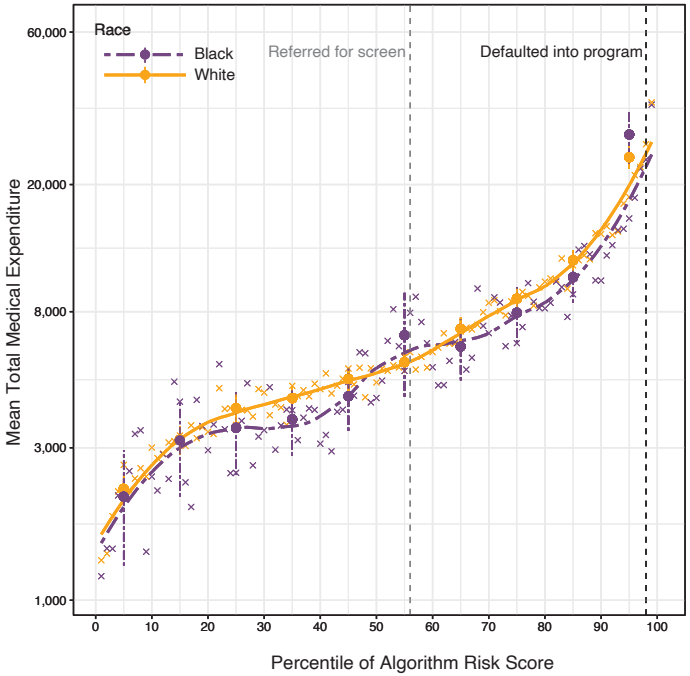
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Understanding Obermeyer et al. (2019) with correlations

- Based on anticipated healthcare expenses, Black and white patients were screened in at equal rates. But there has been historic and ongoing disinvestment for Black patients versus equally sick white patients, which the model picked up and reproduced
- This is exactly the same as actuarial science!



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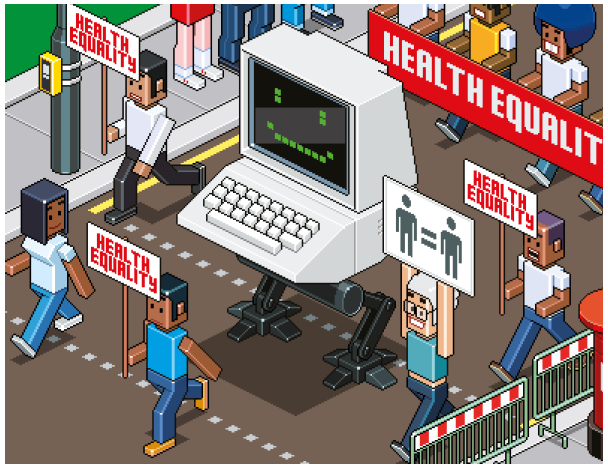
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Thread 1 + 3: ML and healthcare



EQUALITY

Mind the gap

Artificial intelligence has great potential to improve health care, but without careful implementation, only a privileged portion of society will benefit.

BY LINDA NORDLING

When data scientists in Chicago, Illinois, set out to test whether a machine-learning algorithm could predict how long people would stay in hospital, they thought that they were doing everyone a favour. Keeping people in hospital is expensive, and if managers knew which patients were most likely to be eligible for discharge, they could move them to the top of doctors' priority lists to avoid unnecessary delays. It would be a win–win situation: the hospital would save money and people could leave as soon as possible.

Starting their work at the end of 2017, the scientists trained their algorithm on patient data from the University of Chicago academic hospital system. Taking data from the previous three years, they crunched the numbers to see what combination of factors best predicted length of stay. At first they only looked at clinical data. But when they expanded their analysis to other patient information, they discovered that one of the best predictors for length of stay was the person's postal code. This was puzzling. What did the duration of a person's stay in hospital have to do with where they lived?

As the researchers dug deeper, they became increasingly concerned. The postal codes that correlated to longer hospital stays were in poor and predominantly African American neighbourhoods. People from these areas stayed in hospitals longer than did those from more affluent, predominantly white areas. The reason for this disparity evaded the team. Perhaps people from the poorer areas were admitted with more severe conditions. Or perhaps they were less likely to be prescribed the drugs they needed.

The finding threw up an ethical conundrum. If optimizing hospital resources was the sole aim of their programme, people's postal codes would clearly be a powerful predictor for length of hospital stay. But using them would, in practice, divert hospital resources away from poor, black people towards wealthy white people, exacerbating existing biases in the system.

"The initial goal was efficiency, which in isolation is a worthy goal," says Marshall Chin, who studies health-care ethics at University of Chicago Medicine and was one of the scientists who worked on the project. But fairness is also important, he says, and this was not explicitly

TOTO VIZMAN

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"when [the data scientists] expanded their analysis to other patient information, they discovered that one of the best predictors for length of stay was the person's postal code. This was puzzling. What did the duration of a person's stay in hospital have to do with where they lived?"

"The finding threw up an ethical conundrum. If optimizing hospital resources was the sole aim of their programme, people's postal codes would clearly be a powerful predictor for length of hospital stay. But using them would, in practice, divert hospital resources away from poor, black people towards wealthy white people, exacerbating existing biases in the system." (Nordling, 2019)

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Rescher (1998): A normative standard of types of "prediction"

Predictive Approaches	Linking Mechanism	Methodology of Linkage
UNFORMALIZED/JUDGMENTAL		
judgmental estimation	expert informants	informed judgment
FORMALIZED/INFERENTIAL		
RUDIMENTARY (ELEMENTARY)		
trend projection	prevailing trends	projection of prevailing trends
curve fitting	geometric patterns	subsumption under an established pattern
circumstantial analogy	comparability groupings	assimilation to an analogous situation
SCIENTIFIC (SOPHISTICATED)		
indicator coordination	causal correlation	statistical subsumption into a correlation
law derivation (nomic)	accepted laws (deterministic or statistical)	inference from accepted laws
phenomenological modeling (analogical)	formal models (physical or mathematical)	analogizing of actual ("real-world") processes with presumably isomorphic model process

Table 1. A reproduction of figure 6.1 from Rescher (1998, p. 88), "A survey of predictive approaches."

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When are correlations okay to use?

- Very few "umbrella" type problems in healthcare (Kleinberg et al., 2014). We don't want to just passively 'predict' readmission (Marafino et al., 2020), or sepsis, or asthma attacks; we want to predict for the purpose of *stopping* them
- Models can pick up correlations and thereby "predict" well in a "rudimentary (elementary)" way, per Rescher, might be based on non-causal correlations; and so are fragile in that they can easily fail in the future, and worse, we cannot use them as a basis for intervention
  - "Risk factors" that are "modifiable" might nevertheless not be effective to *intervene* on (e.g., maybe the damage is already done from smoking)
- Following Wang, Kapoor et al. (2023), we should consider 'predictive optimization' to be "presumptively illegitimate": that is, illegitimate until/unless all of the potential failure points can be addressed in advance

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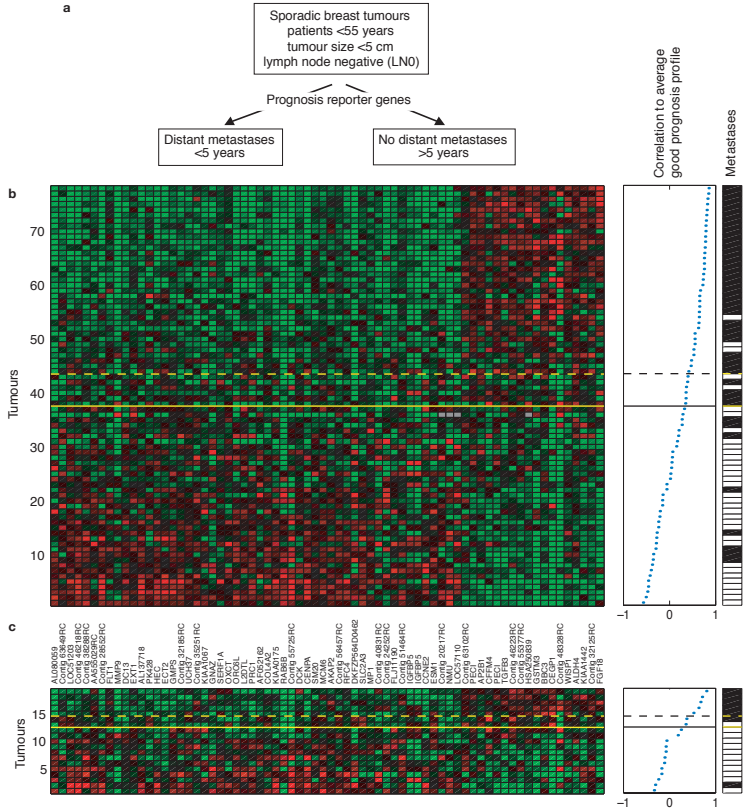
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A positive example



- van't Veer et al. (2002) found 70 genes correlated with developing breast cancer
- Of course the correlations were optimal, post-hoc. But did it generalize?

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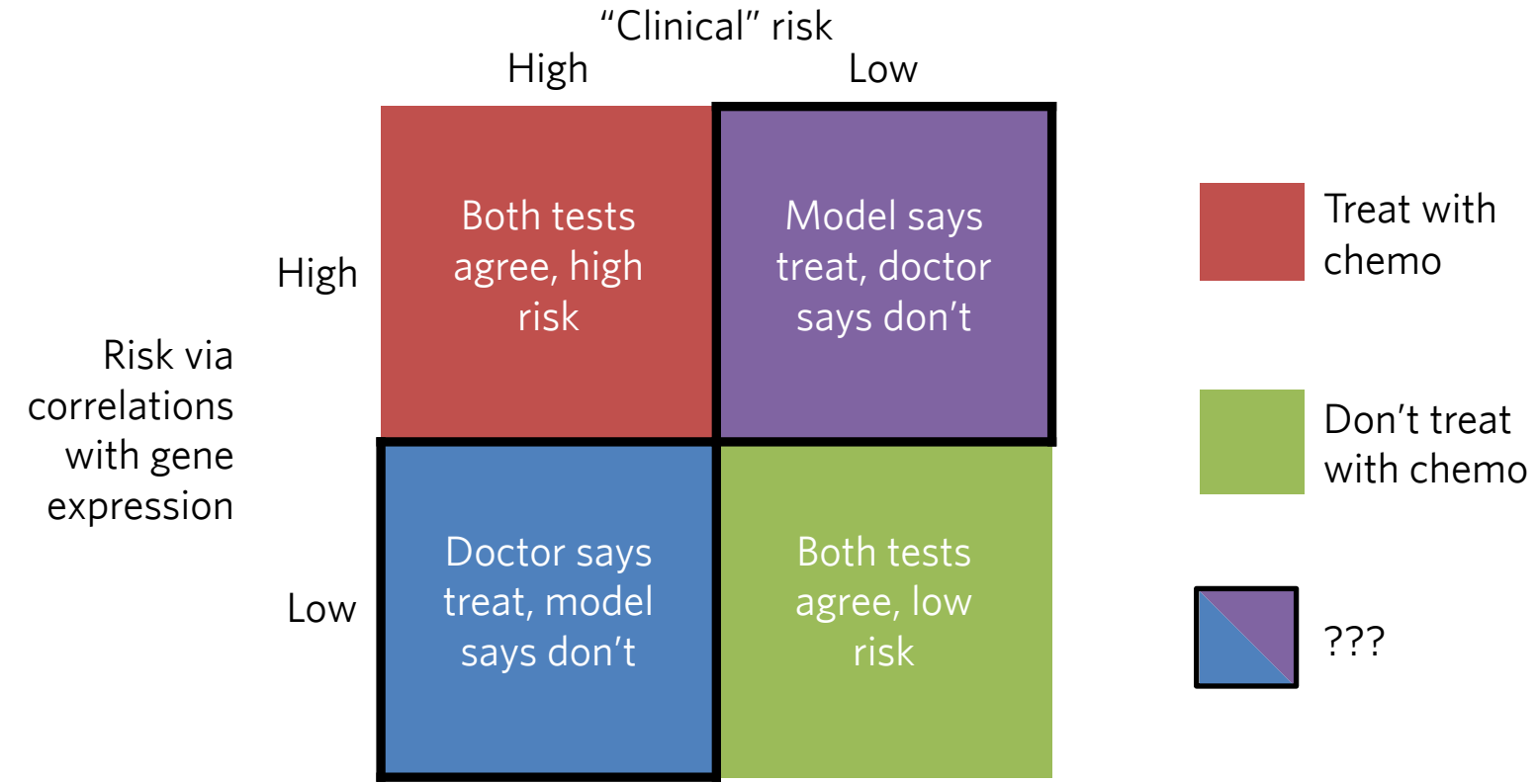
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Implementation testing



Cardoso et al., 2016, *NEJM*

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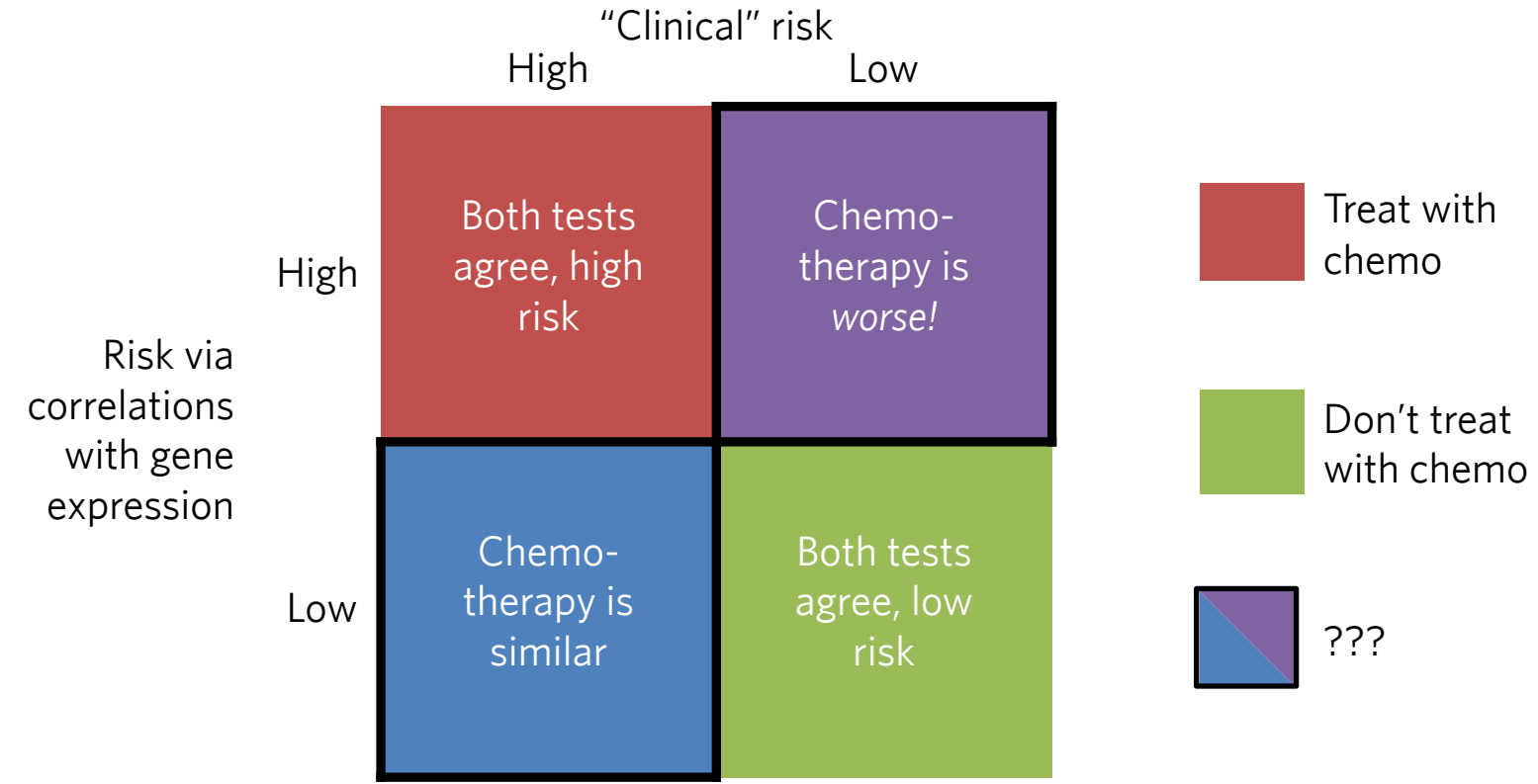
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Cardoso et al., 2016, *NEJM*

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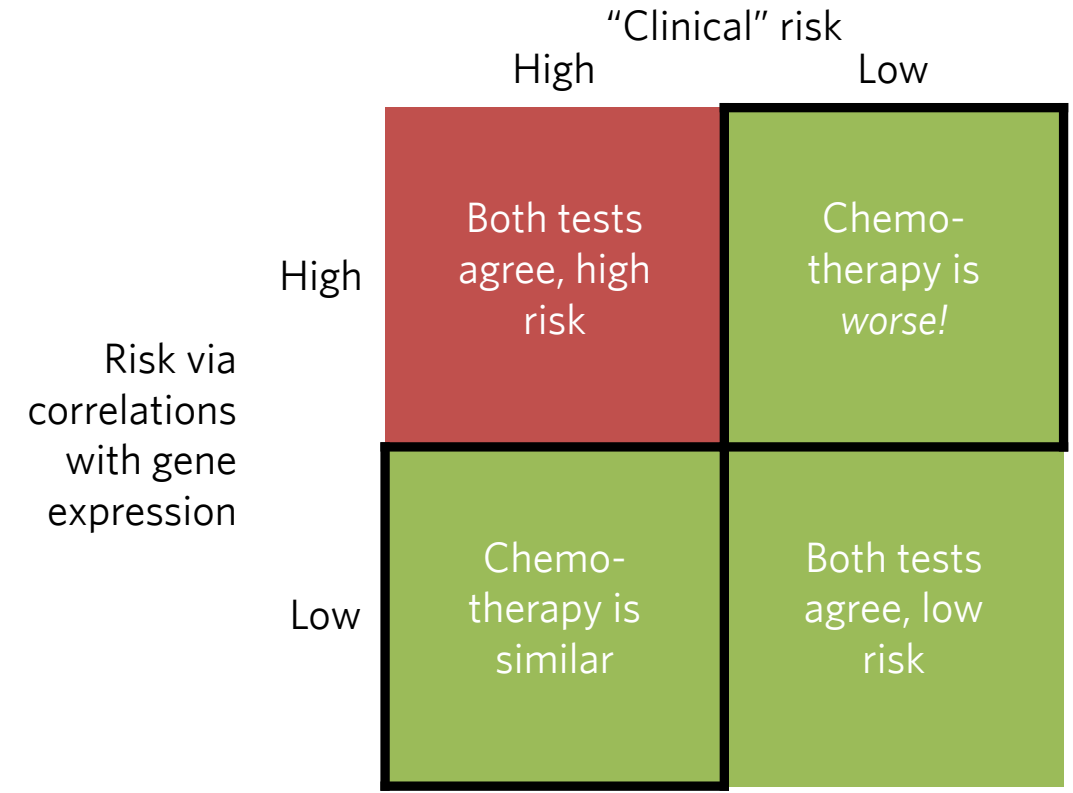
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Implementation testing



Treat with chemo

Don't treat with chemo

Finding: Machine learning *alone* would make things worse. But as a *secondary* diagnosis, on average it catches false positives and avoids unhelpful chemo!

Cardoso et al., 2016, *NEJM*

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A positive example: Lessons

- The question was a causal one: *given a gene signature, will chemotherapy improve outcomes or not*?
- There was no data about this causal question at first; the predictive model was not the right approach, but the only available one
- The RCT (N = 6693) measured and clarified the gap between what was modeled with correlations, and the actual target
- Better would be to re-do the modeling (fit to only N = 98 patients), with modern software and a better model, *on the experimentally collected data*
- The RCT was also not the most representative; better to be able to estimate heterogenous treatment effects

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References (1/3)

Areheart BA. The anticlassification turn in employment discrimination law. Ala Law Rev. 2012;63(5):955-1006. Available from: https://ir.law.utk.edu/utklaw_facpubs/217/.

Angwin J, Larson J, Kirchner L, Mattu S. Minority neighborhoods pay higher car insurance premiums than white areas with the same risk. ProPublica [Internet]. 2017 Apr 5. Available from: https://www.propublica.org/article/minority-neighborhoods-higher-car-insurance-premiums-white-areas-same-risk.

Balkin JM, Siegel RB. The American civil rights tradition: anticlassification or antisubordination? Univ Miami Law Rev. 2003;58(1):9-34. Available from: https://repository.law.miami.edu/umlr/vol58/iss1/5/.

Baiocchi M, Rodu J. Reasoning using data: two old ways and one new. Obs Stud. 2021; 7(1):3-12. doi: 10.1353/obs.2021.0016.

Bouk D. How our days became numbered: risk and the rise of the statistical individual. Chicago: University of Chicago Press; 2015.

Breiman L. Statistical modeling: the two cultures. Stat Sci. 2001 Aug;16(3):199-231. doi: 10.1214/ss/1009213726.

Fergus D. The ghetto tax: auto insurance, postal code profiling, and the hidden history of wealth transfer. In: Harris FC, Lieberman RC, editors. Beyond discrimination: racial inequality in a postracial era. New York: Russel Sage Foundation; 2013. p. 277-316.

Harford T. Big data: a big mistake? Signif. 2014 Dec 1;11(5):14-19. doi: 10.1111/j.1740-9713.2014.00778.x.

He J, Baxter SL, Xu J, Xu J, Zhou X, Zhang K. The practical implementation of artificial intelligence technologies in medicine. Nat Med. 2019 Jan;25(1):30-36. doi: 10.1038/s41591-018-0307-0.

Hellman D. When is discrimination wrong? Cambridge (MA): Harvard University Press; 2008.

Heras AJ, Pradier P-C, Teira D. What was fair in actuarial fairness? Hist Human Sci. 2019 Sep 15;33(2):91-114. doi: 10.1177/0952695119856292.

Ho CS. Commentary on Doe vs. Mutual of Omaha. SSRN: 3635892 [Preprint]. 2020 Jun 26: [20 p.]. Available from: https://ssrn.com/abstract=3635892.

Ho DE, Xiang A. Affirmative algorithms: the legal grounds for fairness as awareness. Univ Chic Law Rev Online. 2020 Oct 30. Available from: https://lawreviewblog.uchicago.edu/2020/10/30/aa-ho-xiang/.

Hoadley B. Comment [on Breiman L, Statistical modeling: the two cultures]. Stat Sci. 2001 Aug;16(3):220-4. doi: 10.1214/ss/1009213726.

Horan C. Insurance era: risk, governance, and the privatization of security in postwar America. Chicago: University of Chicago Press; 2021.

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References (2/3)

Hu L. Disparate causes, pt. II. Phenomenological World. 2019 Oct 7. Available from: https://www.phenomenalworld.org/analysis/disparate-causes-pt-ii/.

Jacobs AZ, Wallach H. Measurement and fairness. Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21); 2021 Mar 3-10. New York: ACM Press; 2021. p. 375-85. doi: 10.1145/3442188.3445901.

Jones ML. How we became instrumentalists (again): data positivism since World War II. Hist Stud Nat Sci. 2018;48(5):673-84. doi: 10.1525/hsns.2018.48.5.673.

Keswani V, Celis LE. An anti-subordination approach to fair classification. SSRN: 4116835 [Preprint]. 2022 May 1: [27 p.]. Available from: https://ssrn.com/abstract=4116835.

Kiviat B. The moral limits of predictive practices: The case of credit-based insurance scores. Am Sociol Rev. 2019;84(6):1134-58. doi: 10.1177/0003122419884917.

Kiviat B. Which data fairly differentiate? American views on the use of personal data in two market settings. Sociol Sci. 2021 Jan 13;8(2):26-47. doi: 10.15195/v8.a2.

Kleinberg J, Ludwig J, Mullainathan S, Obermeyer Z. Prediction policy problems. Am Econ Rev. 2015;105(5):491-5. doi: 10.1257/aer.p20151023.

Landes X. How fair is actuarial fairness? J Bus Ethics. 2015;128:519-33. doi: 10.1007/s10551-014-2120-0.

Malik M, Malik MM. Critical technical awakenings. J Social Comput. 2021 Dec; 4(4):365-384. doi: 10.23919/JSC.2021.0035.

Malik MM. A hierarchy of limitations in machine learning. arXiv: 2002.05193 [Preprint]. 2020 Feb 29: [68 p.].

McDermott D. Artificial intelligence meets natural stupidity. ACM SIGART Bulletin. 1976 Apr;57:4-9. doi: 10.1145/1045339.1045340.

Montgomery B. Wikipedia, vital to AI, takes pains to highlight human contributions. In: Met sacrifices a heap of money at the alter of AI. Guardian. 2025 Jun 17. Available from: https://www.theguardian.com/technology/2025/jun/16/meta-ai-wikipedia-apple-iphone.

Meng XL. Statistical paradises and paradoxes in big data (I): Law of large populations, big data paradox, and the 2016 US presidential election. Ann Appl Stat. 2018 Jun;12(2):685-726. doi: 10.1214/18-AOAS1161SF.

Mitchell S, Potash E, Barocas S, D'Amour A, Lum K. Algorithmic fairness: choices, assumptions, and definitions. Annu Rev Stat Appl. 2021;8:141-63. doi: 10.1146/annurev-statistics-042720-125902.

Mitchell S, Potash E, Barocas S, D'Amour A, Lum K. Prediction-based decisions and fairness: a catalogue of choices, assumptions, and definitions. arXiv: 1811.07867 [Preprint]. 2020 Apr 24: [22 p.]. Available from: https://arxiv.org/abs/1811.07867.

Monahan, J. The clinical prediction of violent behavior. Northvale, NJ: Jason Aronson Inc; 1995. [Originally published in 1981 as Predicting violent behavior: An assessment of clinical techniques by Sage Publishing]

Mullainathan S, Spiess J. Machine learning: An applied econometric approach. J Econ Perspect. 2017;31(2):87-106. doi: 10.1257/jep.31.2.87.

Nopper TK. Digital character in "the scored society": FICO, social networks, and competing measurements of creditworthiness. In: Benjamin R, editor. Captivating technology: race, carceral technoscience, and liberatory imagination in everyday life. Durham (NC): Duke University Press; 2019. p. 170-87. doi: 10.1215/9781478004493-010.

Nordling L. A fairer way forward for AI in health care. Nature. 2019 Sep;573(7775):S103-S105. doi: 10.1038/d41586-019-02872-2.

Norvig P. On Chomsky and the to cultures of statistical learning. 2010. Available from: https://norvig.com/chomsky.html.

Outline

Background

Thread 1: ML to statistics

Thread 2: Injustices of actuarial science

Thread 3: Actuarial logics in healthcare

Tying the threads together

The normative argument

References

References (3/3)

O'Mahony S. Medicine and the McNamara fallacy. J R Coll Physicians Edinb. 2017 Sep;47(3):281-287. doi: 10.4997/JRCPE.2017.315.
Obermeyer Z, Nissan R, Stern M, Eaneff S, Bembeneck EJ, Mullainathan S. Algorithmic bias playbook. Chicago: University of Chicago Booth School of Business, Center for Applied AI; 2022 June. Available from: https://www.chicagobooth.edu/research/center-for-applied-artificial-intelligence/research/algorithmic-bias/playbook.
Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. Science. 2019 Oct 25;366(6464):447-453. doi: 10.1126/science.aax2342.
Ochigame R. The long history of algorithmic fairness. Phenomenal World [Internet]. 2020 Jan 30. Available from: https://www.phenomenalworld.org/analysis/long-history-algorithmic-fairness.
Prince AER, Schwarcz D. Proxy discrimination in the age of artificial intelligence and big data. Iowa Law Rev. 2020;105(3):1257-1318. Available from: https://ilr.law.uiowa.edu/print/volume-105-issue-3/proxy-discrimination-in-the-age-of-artificial-intelligence-and-big-data.
Rescher N. Predicting the future: an introduction to the theory of forecasting. Albany (NY): SUNY Press; 1998.
Roberts JL, Leonard EW. What is (and isn't) healthism. Georgia Law Rev. 2016 Apr 1;50:833-907. Available from: https://digitalcommons.law.uga.edu/fac_artchop/1115.
Roberts JL. "Healthism": a critique of the antidiscrimination approach to health insurance and health-care reform. Univ Ill Law Rev. 2012 Jun 30;4:1159-1206.
Roberts JL. Health law as disability rights law. Minn Law Rev. 2013;97:1963-2035. Available from: https://scholarship.law.umn.edu/mlr/374.
Rodamar J. There ought to be a law! Campbell versus Goodhart. Signif. 2018 Dec;15(6):9. doi: 10.1111/j.1740-9713.2018.01205.x.
Rosenbaum S. An overview of discrimination practices, federal law, and federal reform options. J Law Med Ethics. 2009;37(52):101-20. doi: 10.1111/j.1748-720X.2009.00423.x.
Rothstein WG. Public health and the risk factor: A history of an uneven medical revolution. Rochester (NY): University of Rochester Press; 2008.
Sawhney TG, Pyenson B. When do cause and effect matter for health actuaries? Health Watch [Society of Actuaries Health Section]. 2015 May;78:27-31. Available from: https://www.soa.org/Library/Newsletters/Health-Watch-Newsletter/2015/may/hsn-2015-iss78.pdf.
Selbst AD, boyd d, Freidler SA, Venkatasubramanian S, Vertesi J. Fairness and abstraction in sociotechnical systems. Proceedings of the 2019 ACM Conference on Fairness, Accountability, and Transparency (FAT* '19); 2019 Jan 29-31. New York: ACM Press; 2019. p. 59-68. doi: 10.1145/3287560.3287598.
Shmueli G. To explain or to predict? Stat Sci. 2010 Aug;25(3):289-310. doi: 10.1214/10-STS330.
Silva S, Kenney M. Algorithms, platforms, and ethnic bias. Commun ACM. 2019 Nov;62(11):37-9. doi: 10.1145/3318157.
Silva S, Kenney M. Algorithms, platforms, and ethnic bias: an integrative essay. Phylon. 2018;55(1&2):9-37. Available from: https://www.jstor.org/stable/26545017.
Simons J, Bhatti SA, Weller A. Machine learning and the meaning of equal treatment. Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society (AIES '21); 2021 May 19-21. New York: ACM Press; 2021. p. 956-966. doi: 10.1145/3461702.3462556. Available from: https://ssrn.com/abstract=3678665.
Starr SB. Evidence-based sentencing and the scientific rationalization of discrimination. Stanford Law Rev. 2014;4:803-72. Available from: https://repository.law.umich.edu/articles/1190.
Stevenson MT, Doleac JL. Algorithmic risk assessment in the hands of humans. SSRN: 3489440 [Preprint]. 2021 Apr 22: [62 p.]. Available from: https://ssrn.com/abstract=3489440.
Turner R. "The way to stop discriminating on the basis of race…" Stanf J Civ Rights Civ Liberties. 2015 Apr 9;11:45-88.
van't Veer LJ, Dai H, van de Vijver MJ, He YD, Hart AA, Mao M, Peterse HL, van der Kooy K, Marton MJ, Witteveen AT, Schreiber GJ, Kerkhoven RM, Roberts C, Linsley PS, Bernards R, Friend SH. Gene expression profiling predicts clinical outcome of breast cancer. Nature. 2002 Jan 31;415(6871):530-6. doi: 10.1038/415530a.
Wang A, Kapoor S, Barocas S, Narayanan A. Against predictive optimization: on the legitimacy of decision-making algorithms that optimize predictive accuracy. SSRN: 4238015 [Preprint]. 2022 Oct 04: [45 p.]. Available from: https://ssrn.com/abstract=4238015.
Weber MC. Disability rights, disability discrimination, and social insurance. Ga State Univ Law Rev. 2009 Apr 1;25(3):575-605. Available from: https://readingroom.law.gsu.edu/gsulr/vol25/iss3/5.
Weinberg L. Rethinking fairness: an interdisciplinary survey of critiques of hegemonic ML fairness approaches. J Artif Intell Res. 2022;74:75-109. doi: 10.1613/jair.1.13196.
Wiggins C, Jones ML. How data happened: a history from the age of reason to the age of algorithms. W. W. Norton; 2023.
Xiang A. Reconciling legal and technical approaches to algorithmic bias. Tenn Law Rev. 2021;88:1-75. Available from: https://ssrn.com/abstract=3650635.